Lecture plan · for the lecturer

Learning as optimisation

AI · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Frame learning as minimising empirical risk.
- Run gradient descent and SGD.
- Diagnose learning-rate problems.

Activity: Steps by hand

1. Warm-up (5 min): Show a 2-D loss contour and ask students to draw the gradient arrow at one point.
2. Derive (15 min): Derive the gradient of squared and logistic loss on the board.
3. Step (15 min): Pairs take five steps on θ^2 with $\eta = 0.1$, 0.5 and 1.2 and plot them.
4. Code (20 min): Implement mini-batch SGD and compare loss curves for two batch sizes.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- At which η did θ^2 diverge, and why?
- Why does SGD jitter?
- How will you assess curve reading?

Exit ticket

- Sketch the loss curve for a learning rate that is too big, and label it.

Try it in your next lecture
Give each pair the loss θ^2 and three learning rates; they take five steps by hand and plot them before you show any code.

Seminar sheet · for students

Learning as optimisation

AI · University · Lecture 1

Name _____ Class _____ Date _____

1 **Loss θ^2, $\theta = 3$, $\eta = 0.1$. Where is θ after one step?**

◯ 2.4 ◯ 2.9 ◯ 0

2 **Take two gradient steps on the loss θ^2 with $\eta = 0.25$, starting at $\theta = 8$.**

After step 1, θ = _____ . After step 2, θ = _____ .

Word bank: 4 · 6 · 0 · 2

3 **For the loss θ^2, which learning rates make gradient descent diverge? Choose all that apply.**

☐ $\eta = 1.5$ ☐ $\eta = 0.4$ ☐ $\eta = 1.1$ ☐ $\eta = 0.9$

4 **The negative gradient always points directly at the minimum of the loss.**

◯ True ◯ False

5 **Match each loss curve to its likely cause.**

Loss falls very slowly and steadily	• • • •	Overfitting
Loss grows, then becomes nan	• • • •	Small mini-batches
Loss falls but jitters up and down	• • • •	Learning rate too large
Training loss low, validation loss rising	• • • •	Learning rate too small

✎ **Where do your students first meet gradient descent? What could they do by hand before they see code?**